

Percentages



1 What does the % symbol mean when placed after a number?

2 Fill in the blanks to make the following statements true:

a 29% is equivalent to 29 out of .

b 53% is equivalent to out of 100.



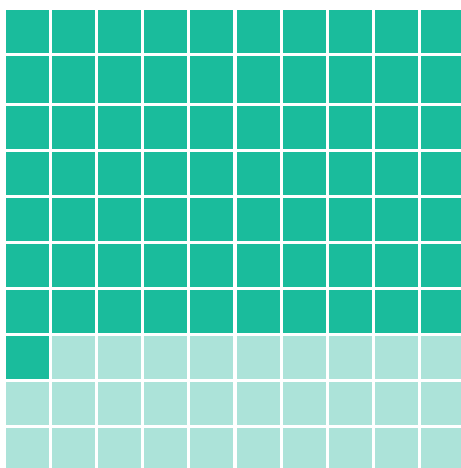
3 Given the grids below, answer the following:

i How many squares are shaded?

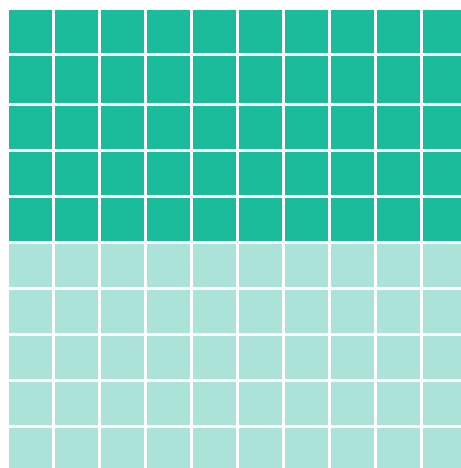
ii What percentage of the grid is shaded?

iii What fraction of the grid does this percentage represent?

a



b



Converting from percentages



4 Write the following as fractions in their simplest form:

a 68%

b 13%

c 55%

d 275%

e 37%

f 94%

g 450%

h 2.5%



5 Write the following as decimals:

a 54%

b 60%

c 270%

d 303%

e 76%

f 500%

g 182%

h 6897%



Converting to percentages

 **6** For each of the following fractions:

- i Write an equivalent fraction with a denominator of 100.
- ii Write it as a percentage.

a $\frac{5}{10}$

b $\frac{9}{300}$

c $\frac{1}{8}$

d $\frac{45}{200}$

7 Write the following as percentages:

a $\frac{83}{100}$

b $1\frac{4}{5}$

c $\frac{42}{100}$

d $\frac{39}{60}$

e $\frac{93}{150}$

f $\frac{21}{35}$

g $3\frac{7}{10}$

h $2\frac{9}{50}$

8 Consider the fraction $\frac{2}{3}$.

- a** Is there an equivalent fraction that has a denominator of 100, and a whole number as its numerator?
- b** What is the closest whole percent to $\frac{2}{3}$?

9 For each of the following pairs, either identify that they are the same or state which number is larger:

a $\frac{1}{8}$, 13%

b $\frac{4}{25}$, 16%

c $\frac{63}{90}$, 70%

d $\frac{1}{3}$, 27%

 **10** Write the following as a percentages:

a 0.24

b 1.1

c 0.857

d 8

e 1.71

f 24

g 13.04

h 0.041

Comparing different values

11 When converting a fraction or decimal to a percentage, why is it necessary to multiply by 100%?

12 When converting from percentage to a fraction or decimal, why is it necessary to divide by



- 13 Convert between percentages, fractions and decimals to complete the table:



Fraction	Decimal	Percentage
$\frac{11}{100}$		
	1.83	
$\frac{5}{8}$		

- 14 Express the following as both a fraction and a decimal:

a 80%

b $47\frac{1}{2}\%$

- 15 For the following groups of numbers:

i Rewrite the group as percentages

ii Hence, order the original group from smallest to largest.

a $\frac{5}{10}$, 100%, 0.3

b $\frac{1}{4}$, 60%, 0.3

c $\frac{3}{8}$, 20%, 0.8

d $\frac{3}{15}$, 45%, 0.27